CaseMatch V1.0.0 User Guide

1 Introduction

CaseMatch is a zero-code, open-source content-based image retrieval (CBIR) software designed for users without programming backgrounds, particularly clinicians and medical researchers. It enables the construction of searchable medical image libraries using state-of-the-art visual encoders through an intuitive graphical interface. The software employs ResNet50 as the core visual feature extractor and provides four pretrained checkpoints, all accessible via a unified GUI.

2 Configuration Files

The software utilizes two configuration files: `config.json` and `model_config.json`.

The `config.json` file contains the following parameters:

```
{
  "common": {
    "batch_size": 16,
    "input_size": 224,
    "resize_before_crop": 256
  }
}
```

The `input_size` and `resize_before_crop` parameters follow the default settings for ResNet50. For smaller images, such as 224×224 H&E histopathology patches, `resize_before_crop` can be adjusted to 224 to avoid unnecessary upscaling.

The default `batch_size` of 16 is optimized for stable execution on both CPU and GPU environments. Users may adjust this value based on available hardware resources.

The `model_config.json` file manages the four integrated ResNet50 checkpoints. The software dynamically loads models by reading this configuration file. Users can incorporate additional supervised or self-supervised domain-specific checkpoints by updating the entries in this file, without modifying the source code.

3 Building an Image Retrieval Library

Creating an image retrieval library involves the following straightforward steps:

1. **Prepare the image collection:** Organize target images in a folder. The software recursively scans all subdirectories to locate supported image files.

2. **Initialize the project:** In the **Build Project** panel, assign a project name, select an available checkpoint, and specify the source image directory.

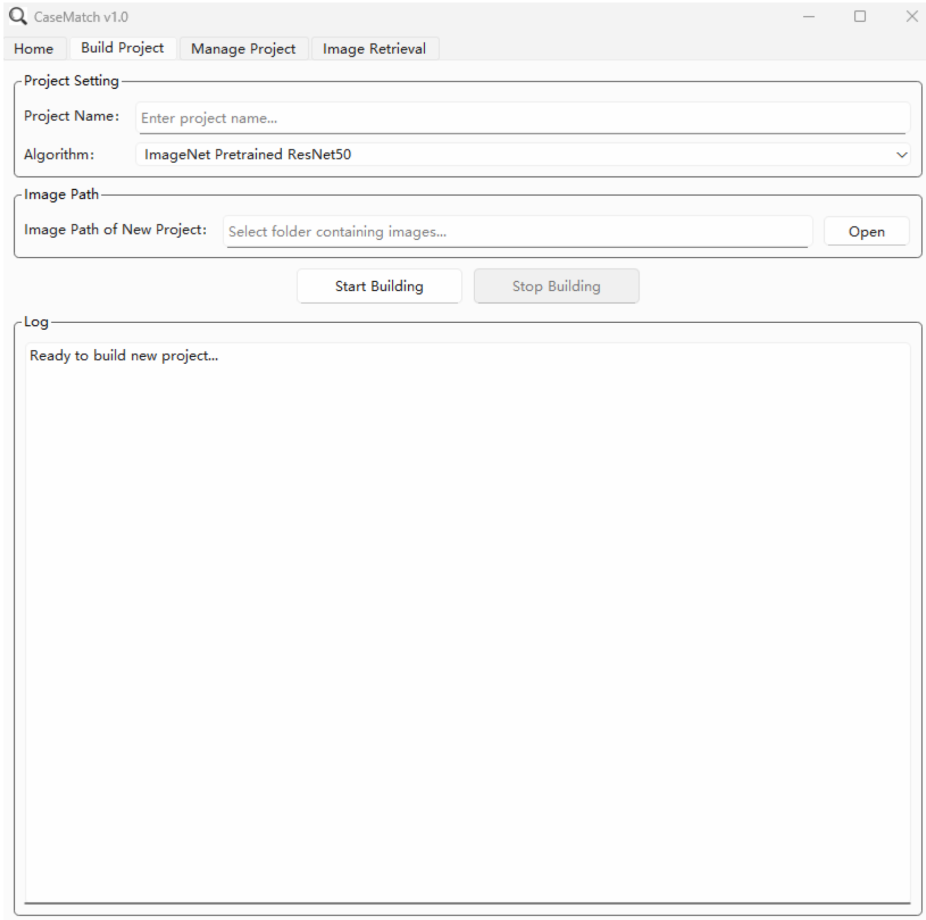

Figure 1. Build Project page.

3. **Vector generation:** The software automatically extracts deep features and constructs the vector index. All metadata and embeddings are stored in the `results` directory under a newly created project folder.

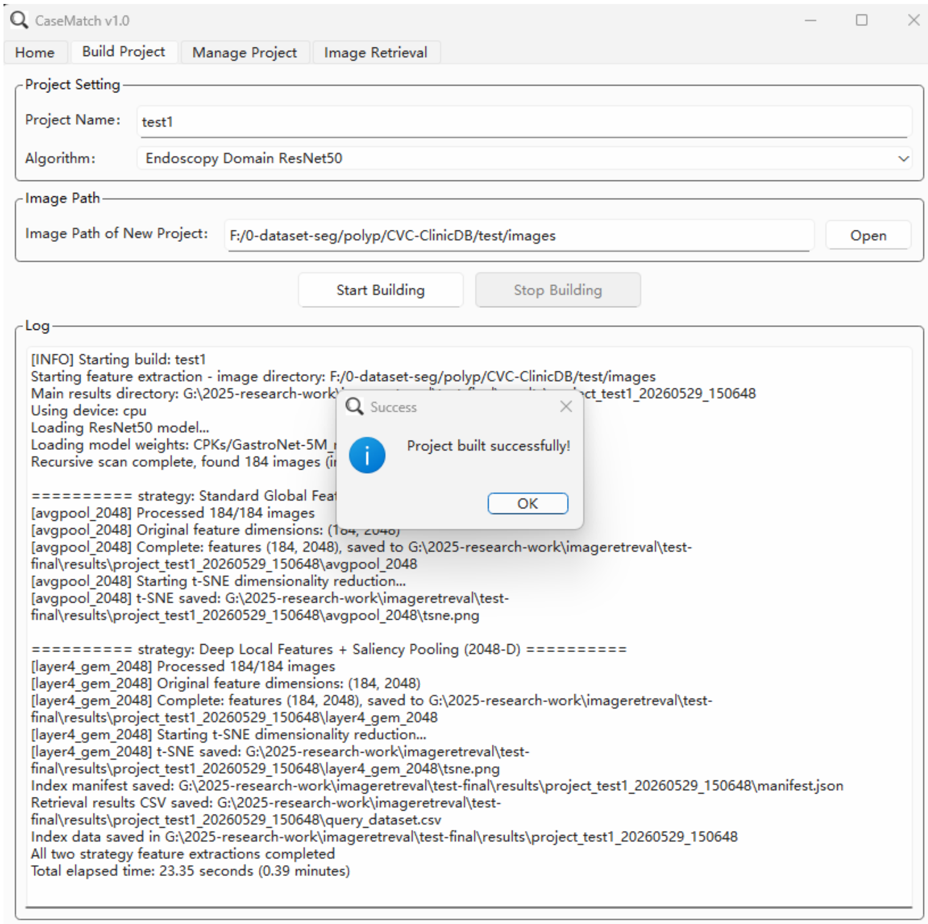

Figure 2. Successful construction of the image vector index library.

4. **Incremental maintenance:** The source image directory remains fixed after initial construction. Users may add or remove images in this directory. In the **Manage Project** panel, select an existing project and trigger a quick rebuild. The system detects newly added or deleted images and updates the index incrementally. Previously indexed images are not reprocessed, ensuring efficient maintenance.

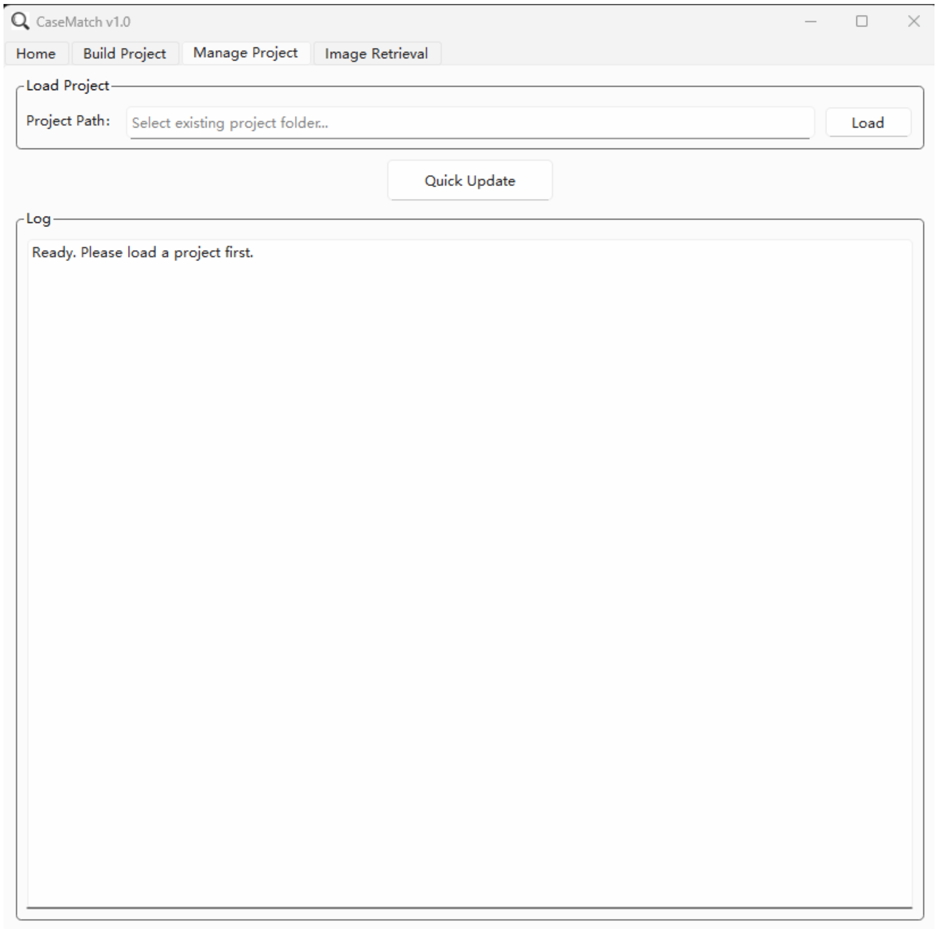

Figure 3. Incremental management for any existing project.

5. **Image retrieval:** In the **Image Retrieval** panel, select an active project, specify a query image path, configure the Top-k retrieval count, and choose the retrieval method. Execute the search to display the Top-5 results in the GUI. Complete retrieval results, including similarity scores and Grad-CAM heatmaps, are exported to the `results` folder.

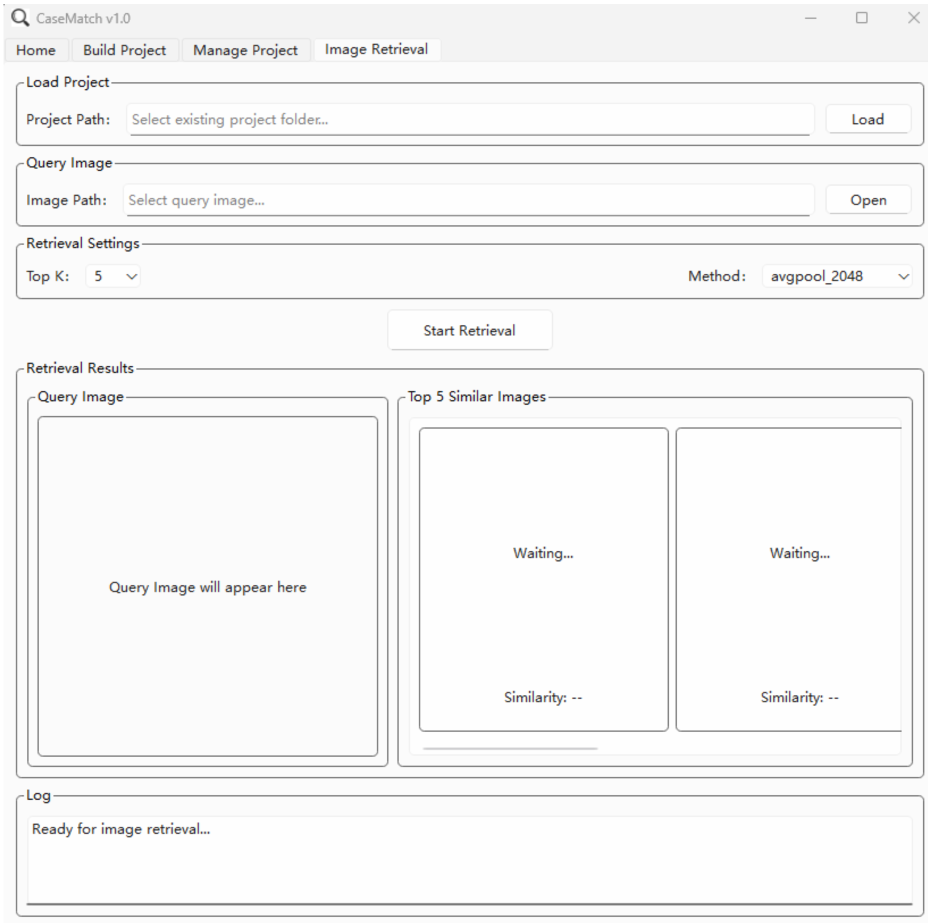

Figure 4. Image Retrieval page.

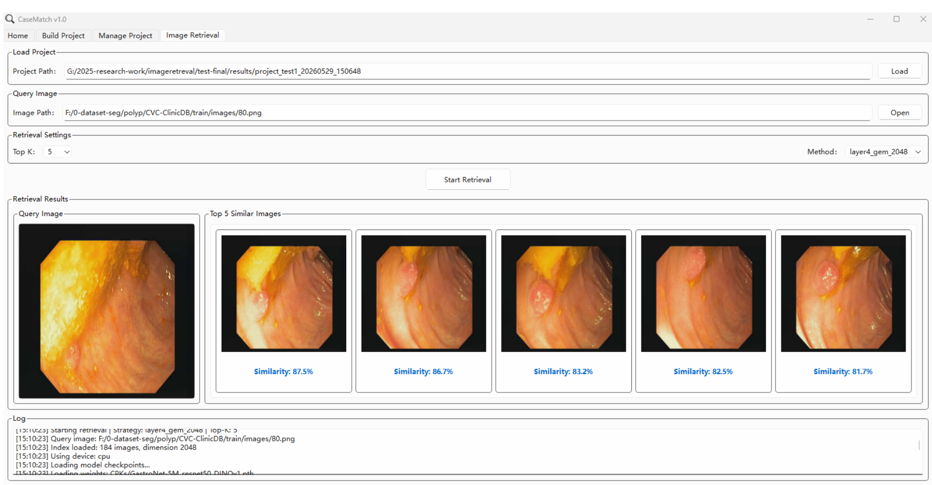

Figure 5. Retrieval of visually similar cases from the library based on a query image. In this example, multiple frames from the same anatomical region of a single patient are successfully retrieved as a contiguous sequence.

4 Software Results

The `results` folder contains two categories of outputs: project management files and retrieval outputs.

4.1 Project Management Files

Each project folder stores the configuration file and computed vector embeddings, as shown below:

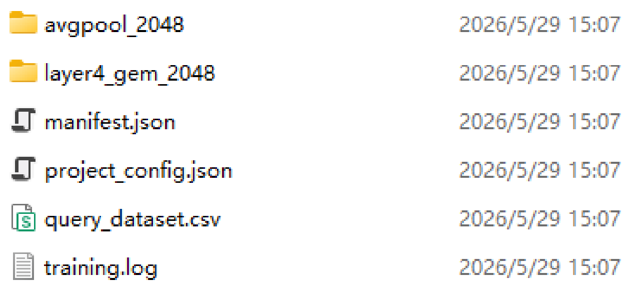

Figure 6. Project result directory containing configuration and vector index files.

4.2 Retrieval Outputs

Retrieval results are organized in a dedicated output folder, as illustrated below:

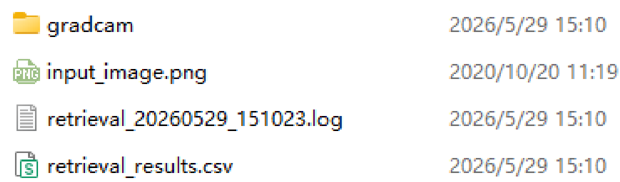

Figure 7. Retrieval result directory structure.

The `gradcam` subdirectory contains detailed activation information for each Top-*k* retrieved image:

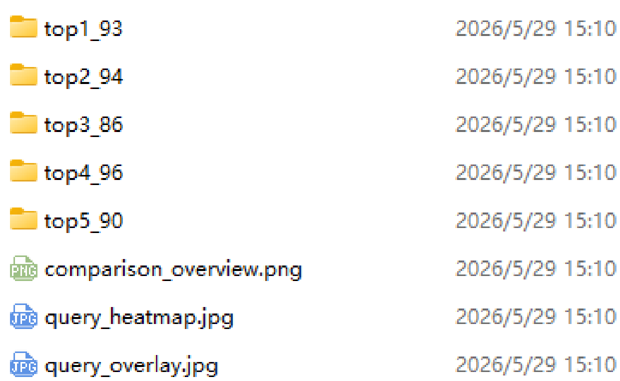

Figure 8. Grad-CAM activation maps for individual retrieved images.

The `comparison_overview` directory provides a consolidated visualization combining the query image, retrieved similar cases, and their corresponding Grad-CAM activation maps:

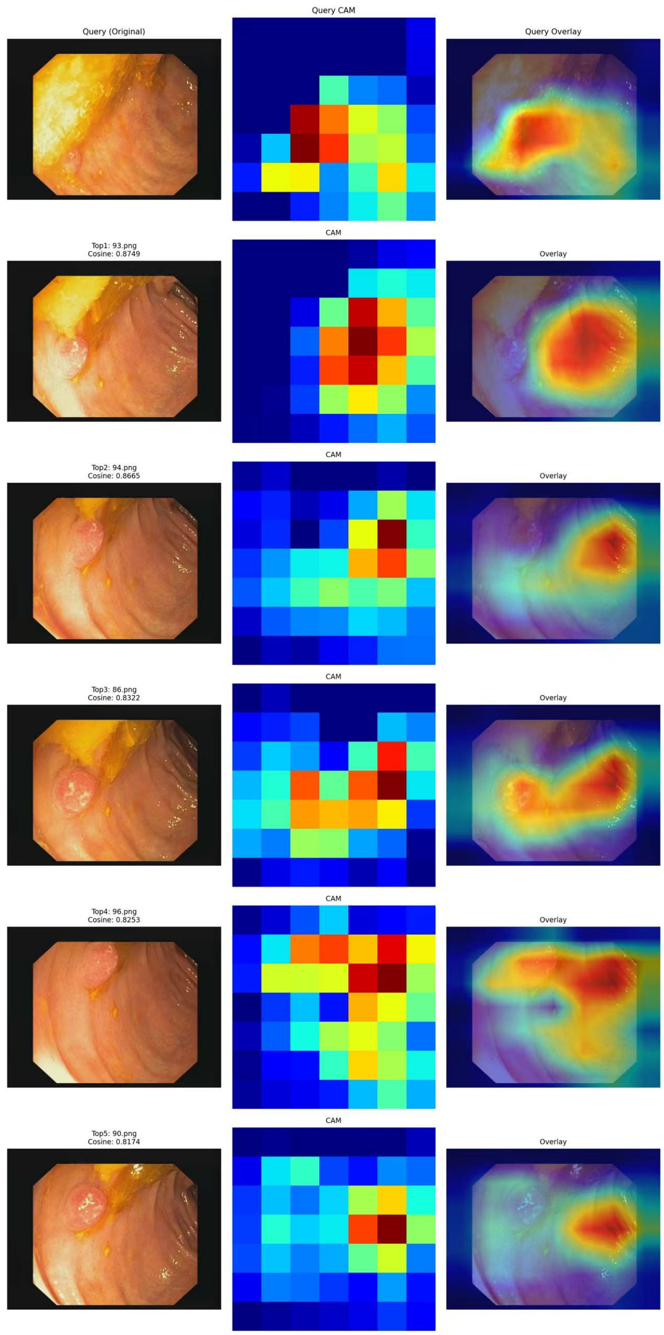

Figure 9. Comprehensive comparison overview showing the query image alongside retrieved similar cases and their Grad-CAM heatmaps.

5 Important Notes

1. **Vector storage:** The software maintains image embeddings in NumPy (`.npy`) format, along with corresponding file lists and original image paths. Each image is represented as a 2048-dimensional feature vector extracted from ResNet50.

2. **Image path integrity:** The software does **not** store original image files. Users must ensure the stability of source image paths and filenames. Any renaming of image files or restructuring of directory paths will invalidate existing references, necessitating a full reconstruction of the retrieval library.

3. **Extensible model configuration:** Users can extend the software's capabilities by updating the `model_config.json` file to register new ResNet50-based visual extractors. Future releases will integrate additional domain-specific foundation

models (e.g., for X-ray and CT imaging) as validated pretrained weights become publicly available.

6 Conclusion

All code is open-source and licensed under the Apache 2.0 license. If you have any questions or need further assistance, feel free to contact me at weihaomeva@163.com. For those who are interested in custom development or support, I am available for paid consultations.

The code repository is available at: https://github.com/AI-thpremed/CaseMatch